Path Ramsey Thresholds in Coprime Graphs

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Abstract

Let G_n be the graph on $\{1, \dots, n\}$ in which two distinct vertices are adjacent when they are coprime. Write $P_{\mathrm{cop}}(k)$ for the least n such that every two-coloring of $E(G_n)$ contains a monochromatic path with k edges. We prove that $P_{\mathrm{cop}}(9) = 17$. An elementary coloring gives the lower bound. For the upper bound, explicit path certificates force any coloring avoiding a nine-edge path to have a connected, 48-edge color graph on 17 vertices in which every four-vertex deletion leaves at least five edges. We classify the possible graphs and show that each admits a four-vertex deletion leaving at most one edge. We also prove

$$2 \leqslant \liminf_{k\to\infty} \frac{P_{\mathrm{cop}}(k)}{k} \leqslant \limsup_{k\to\infty} \frac{P_{\mathrm{cop}}(k)}{k} \leqslant \frac{197\pi^2}{595},$$

and give independently checkable certificates for the previously reported values $P_{\mathrm{cop}}(7) = P_{\mathrm{cop}}(8) = 13$.

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1 Introduction

How large must the coprime graph be before every red–blue edge coloring contains a monochromatic path of a prescribed length? Let G_n have vertex set $[n] = \{1, \dots, n\}$ and edge set

$$E(G_n) = \Big\{\{a, b\} : 1 \leqslant a < b \leqslant n,\ \gcd(a, b) = 1\Big\}.$$

We write $P_{\mathrm{cop}}(k)$ for the least n that forces a monochromatic path with k edges. Throughout the paper, P_j denotes a path on j *vertices*; thus the target defining $P_{\mathrm{cop}}(k)$ is P_{k+1}. All paths are simple. Each color graph has the full vertex set of the host, including any isolated vertices.

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Parity gives a useful starting point. The even vertices of G_n are independent, so the odd vertices form a vertex cover. Partitioning that cover into two small sets gives an edge coloring in which both colors have small vertex covers and therefore no long paths. On G_{16}, this construction avoids a monochromatic nine-edge path. Our main result shows that it cannot be extended to G_{17}.

Theorem 1. *Every two-coloring of $E(G_{17})$ contains a monochromatic P_{10}, whereas some two-coloring of $E(G_{16})$ contains none. Consequently, $P_{\mathrm{cop}}(9) = 17$.*

The upper bound combines three finite properties of G_{17} with an extremal graph argument. The finite properties are certified by explicit paths. They reduce a hypothetical avoiding coloring to two connected color graphs with 47 and 48 edges, each retaining at least five edges after any four vertices are deleted. The following theorem gives the opposing conclusion for the 48-edge graph.

We use $X \vee Y$ for the join of disjoint graphs X and Y, and $\overline{K}_t$ for an independent set of order t.

Theorem 2. *Let J be a connected P_{10}-free graph on 17 vertices with 48 edges. Either J has a vertex cover of size at most four, or*

$$J \cong K_3 \vee (K_3 \cup \overline{K}_{11}).$$

In either case, some four-vertex deletion leaves at most one edge.

The proof of Theorem 2 uses the stability theorems of Füredi, Kostochka, and Verstraëte [6] and Füredi, Kostochka, Luo, and Verstraëte [7], together with the longest-path classification of Ali and Staton [1] and Hamiltonian closure. The graph lies exactly at the edge-count threshold of the relevant stability result, so a separate analysis of equality is needed.

For general path lengths, the same vertex-cover construction gives a lower bound. An application of the Erdős–Gallai theorem to a suitable induced subgraph gives an upper bound.

Theorem 3. *For $k \geqslant 3$,*

$$P_{\mathrm{cop}}(k) \geqslant 4 \left\lfloor \frac{k-1}{2} \right\rfloor + 1.$$

Moreover,

$$2 \leqslant \liminf_{k\to\infty} \frac{P_{\mathrm{cop}}(k)}{k} \leqslant \limsup_{k\to\infty} \frac{P_{\mathrm{cop}}(k)}{k} \leqslant \frac{197\pi^2}{595} = 3.267751373\ldots.$$

In particular, $P_{\mathrm{cop}}(k) = \Theta(k)$.

The path problem differs from the corresponding problem for cliques. Du, Xi, Deng, and Ma [4] express the edge-coloring thresholds for complete targets in terms of classical Ramsey numbers and prime labels. For paths, pulling a coloring back through such labels

can identify distinct vertices, so the same argument does not preserve the required path. Their Table 12 records earlier path computations from the author's repository, and their Open Problem 9.2 asks about non-complete edge targets.

Section 2 gives the Ramsey argument for Theorem 1, using Theorem 2. We prove the classification in Section 3 and the linear bounds in Section 4. Section 5 records certificates for the earlier values $P_{\mathrm{cop}}(7) = P_{\mathrm{cop}}(8) = 13$. The final section discusses the remaining questions, and Appendix A gives the computational evidence and checking instructions.

2 Proof of the exact threshold

2.1 A coloring on sixteen vertices

If a graph has a vertex cover of size m, every path has at most $2m$ edges: it uses at most m vertices of the cover and at most $m+1$ vertices outside it. We will use this observation both here and in the general lower bound.

In G_{16}, partition the odd vertices into

$$A = \{1, 3, 5, 7\}, \qquad B = \{9, 11, 13, 15\}.$$

Color an edge red if it meets A, and blue otherwise. Then A covers the red graph. Since there are no even–even edges, every blue edge meets B, so B covers the blue graph. Both colors therefore have paths of at most eight edges. This proves $P_{\mathrm{cop}}(9) > 16$.

2.2 Three properties of the host

For disjoint sets $A, B \subseteq [17]$, let $G_{17}[A, B]$ denote the graph consisting of the host edges between A and B. We need the following three properties. They will force large color components, prevent sparse remainders after vertex deletions, and rule out the final obstruction to connectedness.

Lemma 4. *The graph G_{17} has the following properties.*

1. *If $A \cup B = [17]$ is a partition with $|A|, |B| \geqslant 5$, then $G_{17}[A, B]$ contains a P_{10}.*
2. *For every four-element set $A \subset [17]$ and every set F of at most four edges of $G_{17} - A$, the graph $G_{17} - A - F$ contains a P_{10}.*
3. *If $|A| = 4$, $e(G_{17}[A]) \geqslant 5$, and $B = [17] \setminus A$, then, for every edge $e \in E(G_{17}[B])$, the graph with edge set $E(G_{17}[A, B]) \cup \{e\}$ contains a P_{10}.*

Proof by finite certificates. For parts 1 and 3, the archive supplies a path for every required cut or pair (A, e). For part 2, it supplies, for each four-vertex deletion up to checked host symmetries, a family of paths that cannot all be met by four edges. Hence at least one path survives any further deletion of at most four edges.

An independent checker verifies every path, the coverage of all required cases, and the symmetries used to reduce the vertex deletions. The path-family condition is checked by exhaustive branching over possible sets of four edges. Appendix A describes this verification and identifies the certificates and checker. □

2.3 From the host properties to connected color graphs

We first recall the exceptional graph in the connected-path bounds. The graph $H(n,k,a)$ has three parts A, B, C of sizes $a, n-k+a, k-2a$, respectively. It has all edges within $A \cup C$ and all edges between A and B, and no others. Thus

$$h(n,k,a) := e(H(n,k,a)) = \binom{k-a}{2} + a(n-k+a). \tag{1}$$

In particular, deleting the four-vertex part A from $H(s,10,4)$ leaves one edge and isolated vertices.

For a connected 13-vertex P_{10}-free graph, Balister, Győri, Lehel, and Schelp [2, Theorem 1.3] give

$$e \leqslant \max\left\{\binom{8}{2} + 5, \binom{5}{2} + 4 \cdot 8\right\} = 42. \tag{2}$$

For a connected P_{10}-free graph of order $s \geqslant 14$, the path stability theorem [6, Theorem 1.6] gives $e \leqslant 3s-3$, unless the graph is a subgraph of $H(s,10,4)$.

Proposition 5. *If a two-coloring of G_{17} has no monochromatic P_{10}, both color graphs are connected on all 17 vertices, their edge counts are 47 and 48, and each retains at least five edges after every four-vertex deletion.*

Proof. We first establish the deletion condition. Suppose that deleting a four-element set A leaves at most four red edges. Delete those edges as well. Every remaining edge is blue, so Lemma 4(2) gives a blue P_{10}, a contradiction. The same argument applies with the colors interchanged.

Next, each color must have a component of order at least 13. Otherwise, a union of its components would have order between 5 and 12. Indeed, if no single component has order in that range, all components have order at most four; accumulate them until the union first has at least five vertices, when it has at most eight. All host edges across the resulting cut have the other color, contrary to Lemma 4(1).

Let $s \geqslant 13$ be the order of the large component in either color. If $s \geqslant 14$, this component cannot be a subgraph of $H(s,10,4)$: deleting its distinguished four vertices would leave at most

$$1 + \binom{17-s}{2} \leqslant 4$$

edges of that color in the whole graph, contradicting the deletion condition. The connected-path bounds therefore give the following upper bounds for the total number of edges in that color:

large component order s	13	14	15	16	17
total edges, at most	48	42	43	45	48.

The first entry is $42 + \binom{4}{2}$; the others are $3s - 3 + \binom{17-s}{2}$. Since $e(G_{17}) = 95$, neither color can have fewer than 47 edges. Thus the two edge counts are 47 and 48, and each large component has order 13 or 17.

It remains to exclude order 13. Suppose a component of one color has vertex set B, with $|B| = 13$, and put $A = [17] \setminus B$. By (2), at least $47 - 42 = 5$ edges of that color lie in A. In particular, $e(G_{17}[A]) \geqslant 5$. Every crossing edge has the other color. If any host edge inside B also had that other color, Lemma 4(3) would give a monochromatic P_{10}. Consequently every host edge inside B has the component's color. But Lemma 4(2), with no edge deletion, gives a P_{10} in $G_{17}[B]$. This contradiction proves that both color graphs are connected on all 17 vertices. □

Proof of Theorem 1. The coloring above proves the lower bound. For the upper bound, suppose that a coloring of G_{17} avoids a monochromatic P_{10}. Proposition 5 supplies a connected, 48-edge color graph in which every four-vertex deletion leaves at least five edges. Theorem 2 supplies a four-vertex deletion leaving at most one edge, a contradiction. □

3 The extremal equality case

We now prove Theorem 2. The argument has three stages. First, a dense bipartite configuration in the universal augmentation of J forces a four-vertex cover. If this configuration is absent, the stability theorems let us delete vertices of degree three until only ten vertices remain. We then classify the possible ten-vertex cores and determine how they can be reconstructed without creating a P_{10}.

3.1 Stability inputs and a forbidden configuration

Adding a universal vertex u to a connected P_{10}-free graph L of order at least two gives a 2-connected graph $L + u$ with no cycle of length at least 11. Indeed, a cycle of that length through u would leave a P_{10} after u is removed, while a cycle avoiding u would itself contain a P_{10}. We call $L + u$ the *universal augmentation* of L.

Let $\mathcal{F}_0$ be the family obtained from $K_{5,6}$ by deleting at most two edges. For a member of this family, write A and B for its parts of sizes five and six. The following specializations of the published stability results will be used. In each statement, Q is 2-connected, has order n, and has circumference $c(Q) < 11$.

1. If $n \geqslant 15$ and $e(Q) > h(n, 11, 4)$, then $Q \subseteq H(n, 11, 5)$ [6, Theorem 1.4].
2. If $n \geqslant 11$ and $e(Q) > \max\{h(n, 11, 4), h(n, 11, 3)\}$, then $Q \subseteq H(n, 11, 5)$ or $Q \subseteq H(n, 11, 2)$ [7, Theorem 2.3].
3. If $n \geqslant 11$, a subgraph of $H(n, 11, 5)$ with more than $h(n, 11, 4)$ edges contains a member of $\mathcal{F}_0$ [7, Lemma 4.6].
4. If an edge contraction Q/xy is 2-connected and contains a member of $\mathcal{F}_0$, then Q contains a member of $\mathcal{F}_0$ [6, Lemma 4.9].

Contraction suppresses loops and parallel edges. The relevant edge counts from (1) are

$$h(n, 11, 4) = 4n - 7, \qquad h(n, 11, 3) = 3n + 4, \qquad h(n, 11, 2) = 2n + 18. \tag{3}$$

We also use the following path property of $\mathcal{F}_0$ [6, Lemma 4.8]. Every pair of distinct vertices in a member of $\mathcal{F}_0$ is joined by a path of at least eight edges. If the pair is not entirely in A, there is such a path of at least nine edges. A path between two vertices of B has even length, so in that case there is a path of ten edges.

Lemma 6. *Suppose that a 2-connected graph Q has circumference less than 11 and contains a member of $\mathcal{F}_0$ with parts A, B. Then $Q - A$ is independent. If $Q = J + u$, where u is universal and $|J| \geqslant 10$, then J has a vertex cover of size at most four.*

Proof. Let S be a component of $Q - (A \cup B)$. It has at least two attachments in $A \cup B$. If $|S| \geqslant 2$, the attachment edges also meet at least two vertices of S; otherwise their unique endpoint in S would be a cut vertex. Thus we can choose attachment edges px, qy with distinct $p, q \in A \cup B$ and distinct $x, y \in S$. A path in S from x to y gives a p–q path of at least three edges, internally outside $A \cup B$. Combining it with a path of at least eight edges in the member of $\mathcal{F}_0$ gives a cycle of length at least 11, a contradiction. Hence every such component S is a singleton.

If a singleton has an attachment in B, join that attachment to a second attachment by a path of at least nine edges in $\mathcal{F}_0$. Together with the two attachment edges, this again gives a cycle of length at least 11. Therefore all vertices outside $A \cup B$ attach only to A. An edge within B would close a ten-edge path in $\mathcal{F}_0$ and is likewise impossible. This proves that $Q - A$ is independent.

In a universal augmentation, u has degree at least ten. Every vertex outside A has degree at most five, so $u \in A$. The set $A \setminus \{u\}$ is consequently a four-vertex cover of J. □

For the rest of the proof, assume that J has no vertex cover of size at most four. By Lemma 6, its universal augmentation contains no member of $\mathcal{F}_0$. The same is true of the augmentation of every induced subgraph of J.

3.2 Deleting vertices down to a ten-vertex core

We first show that every edge of J lies in at least two triangles. The augmentation $J + u$ has 18 vertices and 65 edges. If an edge xy lies in at most one triangle of J, it lies in at most two triangles of $J + u$. Contracting it therefore loses at most three edges, leaving at least 62 edges on 17 vertices. The contracted graph is still a universal augmentation of a connected graph, hence is 2-connected. Contraction cannot increase circumference.

Since $62 > h(17, 11, 4) = 61$, the first and third stability inputs give a member of $\mathcal{F}_0$ after contraction. The fourth input lifts it back to $J + u$, a contradiction. In particular, every vertex of J has degree at least three.

We will preserve the following conditions for a current induced subgraph L of order n:

$$\begin{aligned} &L \text{ is connected and } P_{10}\text{-free}, \qquad e(L) = 3n - 3, \\ &\delta(L) \geqslant 2, \qquad L + u \text{ contains no member of } \mathcal{F}_0. \end{aligned} \tag{4}$$

They hold initially for $L = J$. Suppose that $n \geqslant 11$.

As before, an edge lying in at most one triangle of L can be contracted in $L + u$ at a loss of at most three edges. The augmentation has $n + 1$ vertices and $4n - 3$ edges, so the

contraction has n vertices and at least $4n-6$ edges. This exceeds both $4n-7$ and $3n+4$. The second stability input therefore places the contraction in $H(n,11,5)$ or $H(n,11,2)$. The first alternative is impossible by the dense-subgraph and lifting inputs. For $n \geqslant 13$, the second is also impossible because $4n-6 > 2n+18 = h(n,11,2)$.

For $n = 11, 12$, we exclude the second alternative by counting leaves. Put $M = L/xy$. An old vertex can become a leaf only if it is a common neighbor of x and y, and there is at most one such vertex. The merged vertex can become a leaf only if x and y both had degree two and shared their other neighbor. If this common neighbor also became a leaf, the three vertices would form an isolated triangle in L, contrary to connectedness and $n > 3$. Thus M has at most one leaf.

On the other hand, if $M + u \subseteq H(n,11,2)$, the universal vertex u must lie in the two-vertex joining part of the container. Removing u would make M a connected spanning subgraph of

$$K_1 \vee (K_7 \cup \overline{K}_{n-9}).$$

Each of its $n-9 \geqslant 2$ independent vertices must then be a leaf, a contradiction. We have proved that every edge of L lies in at least two triangles. Hence $\delta(L) \geqslant 3$.

We next exclude $\delta(L) \geqslant 4$. If a longest path in a connected graph is not spanning, it has at least $2\delta+1$ vertices, where δ is the minimum degree. Here is the short argument. Suppose the path is $v_1, \ldots, v_r$ with $r \leqslant 2\delta$. All neighbors of its endpoints lie on the path. Among the $r-1$ indices i, at least δ satisfy $v_1 v_{i+1} \in E(L)$ and at least δ satisfy $v_i v_r \in E(L)$. The two index sets intersect, closing the path into a cycle through all its vertices. Connectivity and a vertex outside the cycle then give a longer path, a contradiction.

In our graph L, a longest path has order $r \leqslant 9 < n$. Thus $\delta(L) \geqslant 4$ would force $\delta(L) = 4$ and $r = 9$. Ali and Staton [1, Theorem 3] classify connected graphs of minimum degree $\delta \geqslant 2$, longest path order $2\delta+1$, and order at least $2\delta+2$. They are subgraphs of $K_\delta \vee \overline{K}_{n-\delta}$ or windmills $K_1 \vee (tK_\delta)$. For $\delta = 4$, a windmill has $5(n-1)/2 < 3n-3$ edges. In the other case, minimum degree four forces every edge between the parts of $K_4 \vee \overline{K}_{n-4}$ to be present. Its augmentation then contains $K_{5,6}$, also impossible. Therefore $\delta(L) = 3$.

Let v be a degree-three vertex. Each edge incident with v lies in two triangles, so the three neighbors of v form a triangle. Deleting v preserves connectedness, removes exactly three edges, and leaves minimum degree at least two. The other conditions in (4) are inherited. We may therefore repeat the deletion.

Seven deletions leave a connected P_{10}-free graph L_{10} on ten vertices with 27 edges and minimum degree at least two. Its predecessor L_{11} had minimum degree at least three. Reversing the process adds one vertex at a time, adjacent to exactly three mutually adjacent vertices already present.

3.3 The possible cores

We show that L_{10} has one of the following three forms:

$$\begin{aligned} &K_2 \vee (K_5 \cup \overline{K}_3), \\ &K_3 \vee (K_3 \cup \overline{K}_4), \\ &\text{a subgraph of } K_4 \vee \overline{K}_6. \end{aligned} \tag{5}$$

Let $Q = L_{10} + u$. It has 11 vertices, 37 edges, and minimum degree at least three, and it has no Hamiltonian cycle. Take its 11-closure: repeatedly add an edge between nonadjacent vertices whose degree sum is at least 11. The Bondy–Chvátal theorem [3] preserves non-Hamiltonicity.

If any edge is added, the closure has at least 38 edges. The two thresholds in the second stability input both equal 37 at $n = 11$, so the closure is a subgraph of $H(11, 11, 5)$ or $H(11, 11, 2)$. The latter has two vertices of degree two and is excluded by the minimum degree. In $H(11, 11, 5) = K_5 \vee \overline{K}_6$, the universal vertex u must lie in the five-vertex part. Removing it gives $L_{10} \subseteq K_4 \vee \overline{K}_6$.

It remains to consider the case where Q is already closed. Write its degrees in nondecreasing order as $d_1, \dots, d_{11}$. Since Q is not Hamiltonian, Chvátal's degree-sequence criterion, in the form stated in [7, Theorem 4.2], gives an index $i < 11/2$ with

$$d_i \leqslant i, \qquad d_{11-i} \leqslant 10 - i.$$

Minimum degree three restricts i to $3, 4, 5$.

For $i = 3$, deleting u gives the following termwise upper bounds for the degrees of L_{10}:

$$(2, 2, 2, 6, 6, 6, 6, 6, 9, 9).$$

Their sum is $54 = 2e(L_{10})$, so equality holds in every position. The two degree-nine vertices are universal. Removing them leaves five vertices of degree four and three isolated vertices, which forces $L_{10} = K_2 \vee (K_5 \cup \overline{K}_3)$.

For $i = 4$, the corresponding upper sequence is

$$(3, 3, 3, 3, 5, 5, 5, 9, 9, 9),$$

again with sum 54. The three degree-nine vertices are universal; removing them leaves a triangle and four isolated vertices. Hence $L_{10} = K_3 \vee (K_3 \cup \overline{K}_4)$.

For $i = 5$, choose a set B of six vertices of degree at most five in Q, and let $A = V(Q) \setminus B$. We will show that an edge inside B would force Q to be Hamiltonian. First, A is a clique. The degree sum of Q is 74, only six below the upper bound

$$6 \cdot 5 + 5 \cdot 10 = 80.$$

If two vertices of A were nonadjacent, closedness would give them degree sum at most ten. Those two vertices alone would contribute a deficit of at least ten, a contradiction.

Let $b = e(Q[B])$, and let m be the number of missing A–B edges. Since A is a clique, the edge count gives $37 = 10 + 30 - m + b$, or $m = b + 3$. For $v \in B$, let s_v count its missing cross edges and t_v its neighbors within B. Its degree is $5 - s_v + t_v \leqslant 5$. Thus, on putting $q_v = s_v - t_v$, we obtain

$$q_v \geqslant 0, \qquad \sum_{v \in B} q_v = m - 2b = 3 - b, \qquad 0 \leqslant b \leqslant 3.$$

For $a \in A$, let r_a count its missing cross edges. Every absent pair av satisfies, by closedness,

$$(10 - r_a) + (5 - q_v) \leqslant 10, \qquad \text{so} \qquad r_a + q_v \geqslant 5. \tag{6}$$

Suppose $b > 0$. Since $q_v \leqslant 3 - b$, every vertex of A incident with a missing cross edge must have at least $b + 2$ such missing edges. There are only $m = b + 3$ missing cross edges in total. Consequently they are all incident with a single vertex $a_0 \in A$. It follows that $s_v \leqslant 1$ for every $v \in B$. As $t_v \leqslant s_v$, the edges in B form a matching.

If $b = 1$, the endpoints of its edge have $q_v = 0$. Equation (6) would require $r_{a_0} \geqslant 5$, whereas $m = 4$. If $b = 2$, the missing cross edges join a_0 to the four matching endpoints and one isolated vertex. If $b = 3$, they join a_0 to all six vertices of B. In both remaining cases,

$$Q \cong K_4 \vee (3K_2 \cup K_1).$$

Indeed, the four vertices of $A \setminus \{a_0\}$ are universal. Outside them there are three edges and one isolated vertex; when $b = 2$, the third edge joins a_0 to its sole neighbor in B.

This graph is Hamiltonian: place its three two-vertex blocks and its singleton between successive vertices of the K_4 in a cycle. Thus $b > 0$ is impossible. We have $b = 0$, and deleting the universal vertex $u \in A$ again gives $L_{10} \subseteq K_4 \vee \overline{K}_6$. This completes the list (5).

3.4 Reconstructing the graph

We now examine which cores can survive the seven reverse additions. Each new vertex has a triangle as its neighborhood among the vertices already present.

The first core, $K_2 \vee (K_5 \cup \overline{K}_3)$, is impossible. Its three independent vertices have degree two, so all three would have to be adjacent to the first new vertex in order to obtain $\delta(L_{11}) \geqslant 3$. But three independent vertices cannot form its required neighborhood triangle.

For the second core, $K_3 \vee (K_3 \cup \overline{K}_4)$, let A be the joining triangle, C the other triangle, and B the independent set. Every vertex outside A is an endpoint of a nine-vertex path, as the following templates show:

$$c_1, c_2, c_3, a_1, b_1, a_2, b_2, a_3, b_3,$$
$$b_1, a_1, c_1, c_2, c_3, a_2, b_2, a_3, b_3.$$

A new vertex cannot be adjacent to such an endpoint, since prepending the new vertex would give a P_{10}. Its earlier neighborhood must therefore be exactly A. The same templates apply to this new vertex as a member of the enlarged independent set B. Induction forces

all seven additions to have earlier neighborhood A. No edges join the added vertices, and the final graph is $K_3 \vee (K_3 \cup \overline{K}_{11})$.

Finally, suppose $L_{10} \subseteq K_4 \vee \overline{K}_6$, with parts A and B. The container has 30 edges and the core has 27, so at most three cross edges are missing. We claim that every vertex of B, and every previously added vertex whose earlier neighborhood lies in A, is an endpoint of a nine-vertex path.

To prove the claim, let z be such a vertex. Choose a neighbor $a_1 \in A$ of z and order the other three vertices as a_2, a_3, a_4. Select five original vertices of B, excluding z if $z \in B$, and call this set B'. We seek distinct representatives in B' for the four slots

$$\{a_1, a_2\}, \qquad \{a_2, a_3\}, \qquad \{a_3, a_4\}, \qquad \{a_4\}.$$

A vertex is eligible for a slot if it is adjacent to every vertex in that slot.

Each slot has at least two eligible vertices, since there are five candidates and at most three missing cross edges. This proves Hall's condition for one or two slots. For a collection of three or four slots, a vertex outside the union of their eligible sets must account for at least two missing cross edges: one missing edge affects at most two slots. With only three missing cross edges in total, at most one vertex of B' can be outside such a union. The union therefore has at least four vertices. Hall's condition holds for every collection of slots.

The resulting distinct representatives give the path

$$z, a_1, b_1, a_2, b_2, a_3, b_3, a_4, b_4,$$

proving the claim. Now consider the first added vertex whose earlier neighborhood is not contained in A. Its earlier neighbor outside A starts a path of this form, so prepending the new vertex creates a P_{10}. No such addition is possible. Thus A covers the final graph, contrary to our standing assumption.

Only the triangle-join graph remains when there is no four-vertex cover. In that graph, deleting the joining triangle and one vertex of the other triangle leaves one edge. In the cover case, deleting four vertices containing the cover leaves no edges. This proves Theorem 2.

4 Linear bounds

Proof of Theorem 3. The lower bound. Put $m = \lfloor (k-1)/2 \rfloor$. The $2m$ odd vertices cover G_{4m}. Partition them into two sets of size m, color every edge meeting the first set red, and color the remaining edges blue. Each color has a cover of size m, so its paths have at most $2m < k$ edges. This gives

$$P_{\mathrm{cop}}(k) \geqslant 4m + 1 = 4 \left\lfloor \frac{k-1}{2} \right\rfloor + 1$$

and hence the asymptotic lower constant two.

A density upper bound. By the Erdős–Gallai path theorem [5], a graph with no k-edge path has at most $(k-1)v/2$ edges. If a two-coloring of a subgraph $H \subseteq G_n$ avoids such a path, applying the theorem to both color graphs gives $e(H) \leqslant (k-1)v(H)$. Therefore

$$e(H) > (k-1)v(H) \quad \Longrightarrow \quad P_{\mathrm{cop}}(k) \leqslant n. \tag{7}$$

The strict inequality in this criterion is essential.

Write $E_n = e(G_n)$. Counting ordered coprime pairs by Möbius inversion gives

$$2E_n + 1 = \sum_{r=1}^{n} \mu(r)\lfloor n/r \rfloor^2 = \frac{6}{\pi^2} n^2 + O(n \log n).$$

The extra one counts the pair $(1,1)$. Replacing the floors by their arguments contributes $O(n \log n)$, and the tail of $\sum_{r \geqslant 1} \mu(r)/r^2 = 1/\zeta(2)$ contributes $O(n)$. Taking $H = G_n$ in (7) already yields

$$\limsup_{k \to \infty} \frac{P_{\mathrm{cop}}(k)}{k} \leqslant \frac{\pi^2}{3}.$$

Improving the edge-to-vertex ratio. To improve the constant, we remove a set of vertices that contributes relatively few edges. We use the multiples of 30, 42, and 70. Their small prime factors restrict their coprime neighbors, and all these vertices are even, so they form an independent set. Put

$$D_n = \{a \leqslant n : 30 \mid a \text{ or } 42 \mid a \text{ or } 70 \mid a\}, \qquad H_n = G_n - D_n.$$

Every pair of the three divisors has least common multiple 210. Inclusion-exclusion gives

$$\begin{aligned} |D_n| &= \lfloor n/30 \rfloor + \lfloor n/42 \rfloor + \lfloor n/70 \rfloor - 2\lfloor n/210 \rfloor \\ &= \frac{13}{210} n + O(1). \end{aligned}$$

We also need to count the edges lost. For a fixed squarefree integer $d \geqslant 2$, let $C_d(n)$ count ordered coprime pairs $(a,b) \in [n]^2$ with $d \mid a$. Möbius inversion gives

$$\begin{aligned} C_d(n) &= \sum_{r=1}^{n} \mu(r) \left\lfloor \frac{n}{\operatorname{lcm}(d,r)} \right\rfloor \left\lfloor \frac{n}{r} \right\rfloor \\ &= \frac{6n^2}{\pi^2 \prod_{p \mid d}(p+1)} + O_d(n \log n). \end{aligned}$$

The floor and tail estimates are as above. The leading coefficient follows from the Euler product

$$\begin{aligned} \sum_{r \geqslant 1} \frac{\mu(r)}{r \operatorname{lcm}(d,r)} &= \frac{1}{d} \prod_{p \mid d} (1 - 1/p) \prod_{p \nmid d} (1 - 1/p^2) \\ &= \frac{6}{\pi^2 \prod_{p \mid d}(p+1)}. \end{aligned}$$

Because D_n is independent, each lost edge is counted exactly once by an ordered pair whose first endpoint is deleted. The number of lost edges is therefore

$$\begin{aligned} &C_{30}(n) + C_{42}(n) + C_{70}(n) - 2C_{210}(n) \\ &\quad = \frac{6}{\pi^2}\left(\frac{1}{72} + \frac{1}{96} + \frac{1}{144} - \frac{2}{576}\right) n^2 + O(n \log n) \\ &\quad = \frac{n^2}{6\pi^2} + O(n \log n). \end{aligned}$$

It follows that

$$v(H_n) = \frac{197}{210} n + O(1), \qquad e(H_n) = \frac{17}{6\pi^2} n^2 + O(n \log n),$$

and hence

$$\frac{e(H_n)}{v(H_n)} = \frac{595}{197\pi^2} n + O(\log n).$$

For any $\varepsilon > 0$, choose

$$n = \left\lceil \left(\frac{197\pi^2}{595} + \varepsilon \right) k \right\rceil.$$

The positive linear margin from εk dominates the logarithmic error, so (7) applies for all sufficiently large k. Letting ε tend to zero proves the asserted upper bound. □

5 Certificates for the earlier thresholds

The values

$$P_{\mathrm{cop}}(7) = P_{\mathrm{cop}}(8) = 13$$

were reported in the author's earlier computations and appear in Table 12 of Du et al. [4]. We record their certificates here for completeness. The upper-bound certificate uses a different method from the explicit host paths at $n = 17$.

For the lower bound, partition the odd vertices of G_{12} into $\{1, 3, 11\}$ and $\{5, 7, 9\}$. The vertex-cover coloring used in Section 2 gives each color a cover of size three, so neither color has a path with more than six edges. Thus $P_{\mathrm{cop}}(7), P_{\mathrm{cop}}(8) > 12$.

For the upper bound at $n = 13$, assign one Boolean variable to each of the 57 host edges. An eight-edge path gives two clauses, excluding its two monochromatic assignments. The archived formula contains 71 832 valid path-exclusion clauses and one harmless tautology ensuring that the solver registers all 57 variables. It imposes no symmetry restrictions.

An independent semantic check verifies that every path clause excludes a monochromatic assignment of a valid eight-edge path. A separate DRAT certificate establishes that the formula is unsatisfiable. This subset of the path constraints is enough: every coloring with no monochromatic eight-edge path would satisfy all of them. Hence every coloring of G_{13} contains an eight-edge path, and therefore also a seven-edge path. Together with the lower coloring, this proves the two previously reported values.

Appendix A identifies the formula, DRAT proof, verification records, and required checkers.

6 Related bounds and further questions

The equality case in Theorem 2 lies just outside the strict stability bounds used in its proof. The universal augmentation has 18 vertices and 65 edges, exactly $h(18, 11, 4)$. The later refinements of Ma and Ning [8] also retain a strict inequality at this point. In their notation, $W_{n,a,c} = H(n, c+1, a)$, and at $(n, c) = (18, 10)$ the term $f(n, 4, c)$ equals 65. Their Theorems 1.7 and 1.9 require more edges than a maximum containing this term, so neither directly gives the boundary classification needed here. This comparison concerns those statements; a comprehensive determination of priority for the classification is outside the scope of the paper.

The lower construction suggests asking whether

$$P_{\mathrm{cop}}(2m+1) = 4m+1 \qquad \text{for every } m \geqslant 1.$$

Theorem 1 proves the case $m = 4$, but its host-specific certificates do not establish the general upper bound. It also remains to close the gap between the asymptotic constants 2 and $197\pi^2/595$. The deletion set used in Section 4 gives an explicit improvement over the full-host density bound; we do not claim that this choice is optimal or that the refinement has priority over all earlier work.

A shorter structural proof of $P_{\mathrm{cop}}(7) = P_{\mathrm{cop}}(8) = 13$ would be useful as well. It would give an account of these earlier values without the separate SAT certificate.

A Certificates and reproducibility

The computational evidence and proof notes are available in the repository at the immutable revision `8bda81e` [9]. This revision predates the present editorial revision. All paths below are relative to the repository root.

A.1 What each archive contains

The three host properties. The structural archive contains the crossing paths, deletion families, and cut extensions certifying the three parts of Lemma 4.

The nine-edge threshold. The resolution archive contains `lower-n16-k9.json`, which records all 79 colored edges of the lower witness, together with source and verification records.

The earlier thresholds. The earlier path archive contains the 45-edge lower witness at $n = 12$, the compressed formula and DRAT proof at $n = 13$, and their verification records.

The first two archives use checkers written in standard-library Python; verifying their certificates requires no SAT solver.

A.2 Verifying the three host properties

For Lemma 4(1), a certificate gives a crossing path for each smaller side of a cut. There are

$$\sum_{s=5}^{8} \binom{17}{s} = 62\,322$$

such sides; taking complements covers the other orientations. For part 3, there is one path for every qualifying pair (A, e).

For part 2, there are $\binom{17}{4} = 2380$ vertex deletions. Checked host automorphisms reduce them to 262 equivalence classes. For each class, the certificate supplies a family of surviving paths with no edge transversal of size at most four. Here an *edge transversal* is a set of edges meeting every path in the family. If no such transversal exists, deleting four edges cannot destroy all the paths.

The checker establishes nonexistence of a transversal by exhaustive branching. It selects a path not yet met by the chosen edges and branches on each of that path's edges as the next choice. Any transversal extending the current choice must contain at least one of these edges, so the branches cover every possibility. Branches requiring more than four edges are rejected.

For every supplied path, the checker verifies ten distinct vertices, all nine gcd adjacencies, and the relevant cut or deletion restrictions. It also verifies every transposition within each declared vertex class against host adjacency. These transpositions generate the within-class permutations, so deletions with the same class counts are equivalent. The checker derives the required cases from the task parameters and requires every deletion signature exactly once. It does not rely on saved completion flags.

The expected counts are:

Certificate	Verified cases or witnesses
Cuts	62 322 cuts
Vertex and edge deletions	2380 deletions in 262 classes
	30 671 paths in the families
Extensions	63 859 pairs (A, e) on 1253 sets A

To obtain the cited revision and run the checks with Python 3.10 or newer:

```
git clone https://github.com/queelius/computational-explorations.git
cd computational-explorations
git checkout --detach 8bda81e8d3f867bb1450b0e492592164416ba98f

cert_dir=docs/audits/coprime-path-nine-2026-10-18
checker=src/coprime_path_structure_certificates.py
python3 -I -S -O "$checker" cuts "$cert_dir/cuts.json.gz"
python3 -I -S -O "$checker" faults "$cert_dir/fault-families.json.gz"
python3 -I -S -O "$checker" extensions "$cert_dir/extensions.json.gz"
```

Require a successful exit, the counts above, and the intended task parameters: $n = 17$, target length nine, minimum cut side five, and vertex- and edge-deletion budgets of four. For extensions, require `side_size=4` and `minimum_side_edges=5`. The symmetry checks concern the host only; no assumption is made that a coloring respects those symmetries.

A.3 Lower colorings and the separate SAT certificate

The complete lower colorings are checked independently by subset dynamic programming. For each vertex subset and color, it records the possible endpoints of a path using exactly that subset. The longest monochromatic path lengths, measured in edges, are $(8, 8)$ at $n = 16$ and $(6, 6)$ at $n = 12$.

For the $n = 13$ upper bound, a separate formula checker verifies that every non-tautological clause has uniform sign and consists of the eight distinct edges of a simple path. The unsatisfiability proof is checked with DRAT-trim, pinned at revision `2e3b2dc0ecf938addbd779d42877b6ed69d9a985`. The archive README gives the complete commands. Its verification record reports a proof core with 57 506 clauses, 391 234 lemmas, and 4 858 014 resolution steps, with no RAT lemmas in the core.

These checks have separate roles: path and coloring checkers validate explicit combinatorial witnesses; the formula checker validates the path constraints; DRAT-trim validates the proof of their inconsistency. The $n = 17$ argument uses the explicit host certificates and the written extremal proof, with no SAT unsatisfiability certificate.

The verification depends on the checkers, their execution environments, and the published mathematical inputs. Checking the finite claims does not formally verify the whole written argument, for which there is no Lean formalization. Unfinished or time-limited searches supply no proof of the statements here.

A.4 Historical records

The earlier code archive is available at doi:10.5281/zenodo.23433640. The earlier manuscript is available at doi:10.5281/zenodo.19058647.

Both deposits are retained as historical records. The new nine-edge-path evidence is identified by the immutable repository revision above, rather than by these earlier deposits.

AI assistance and draft status

Codex assisted with the proof investigation, certificate software, literature checks, and preparation and editorial revision of this manuscript. Human verification of every proof step and citation, and the author's final revision of the AI-assisted proofs in the author's own words, remain pending. This is a working draft and has not been submitted to a journal.

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